Q-Relay: QRNG → Teleportation → Distributed Deutsch–Jozsa

Qiskit Fall Fest 2026 · DBATU, Lonere. All three problem statements, built as one pipeline in which each stage depends on the previous one, with every result labelled by source: **SIM** (Aer simulator), **EMU** (Aer plus the noise model of a real IBM device snapshot, `FakeBrisbane`) or **HW** (real IBM hardware, with job ID).

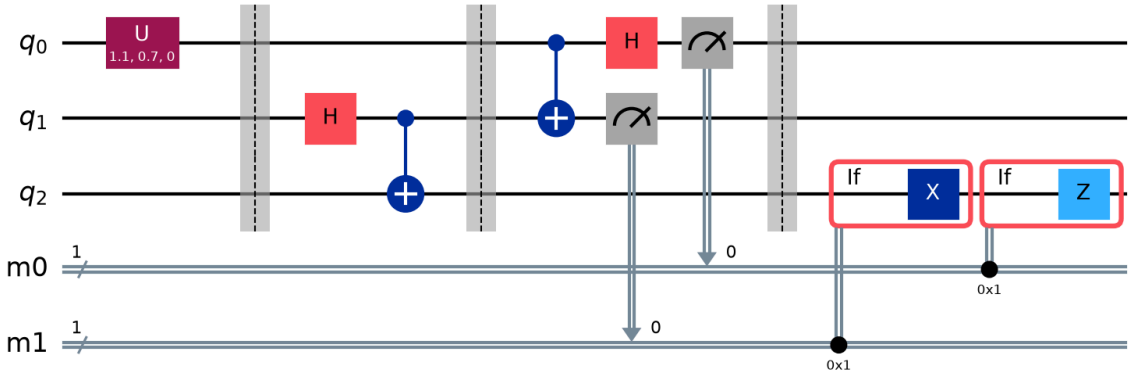
Stack: Python 3.12.5, Qiskit 2.6.1, qiskit-aer 0.18.2, qiskit-ibm-runtime 0.51.0, NumPy 2.5.4, SciPy 1.18.1. Code: `qrelay/` (library), `notebooks/01-04` (executed), `tests/` (53 tests), `web/` + `server/` (interactive demo). Derivations: `docs/THEORY.md`.

> **Honesty notes.** (1) Aer's measurement outcomes come from a classical PRNG, so SIM "quantum" bits are pseudo-random. Only HW bits are physically random. (2) EMU results are never presented as hardware. (3) Passing statistical tests does not prove randomness. (4) A teleportation channel is certified quantum only when average fidelity exceeds the classical bound 2/3. Hardware numbers come from one IBM job (`dbgnmpote5oc738iia1g` , ibm_marrakesh, 2026-10-28).

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1. PS1: Quantum teleportation

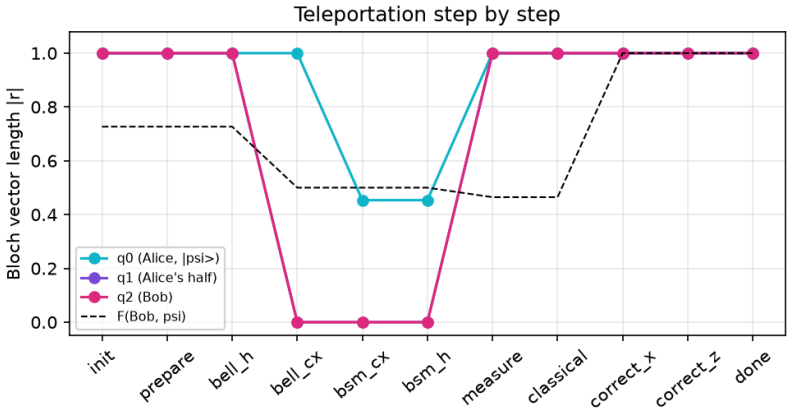
Approach. Alice holds $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$ (q0) and half of a Bell pair (q1). Bob holds q2. The primary implementation is a **dynamic circuit**: mid-circuit Bell-basis measurement and classical feed-forward via `if_test` (X if m_1, then Z if m_0). A **deferred-measurement** variant (CX/CZ corrections) serves as the universal fallback. A third engine evolves the 3-qubit density matrix gate by gate and records every qubit's Bloch vector for the visualisation.



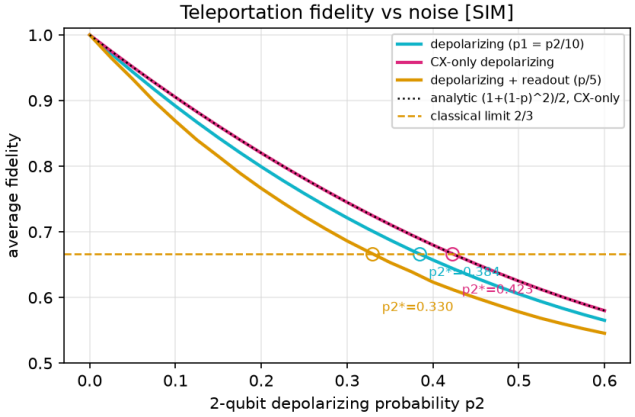
Verification. Four independent measures:

check	result [SIM]
Exact $F = \langle\psi\|\rho_B\|\psi\rangle$ in **each** of the 4 branches (Aer conditional density matrices)	1.000000000 in every branch, for 50 random states, both variants
6 cardinal states, Pauli transfer matrix	$R = I$ (to 1e-9), $F_{pro} = 1$
Tomography (Z/X/Y bases, our implementation), 200 Haar states from the QRNG pool, 1024 shots/basis	$\bar{F} = 0.9916 \pm 0.0050$ (finite-shot spread)
Exact average over the same 200 states	mean 1.0000000000, min 0.9999999999999994

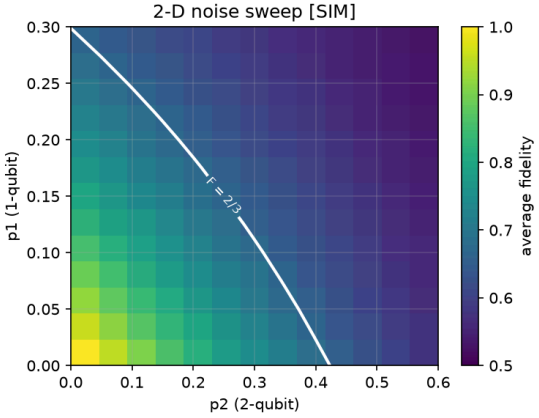
The step engine shows the physics: after the Bell CX both q1 and q2 have $|\vec{r}| = 0$ (maximally mixed halves). After Alice's measurement q0 collapses to a pole (**no-cloning**: moved, not copied). Bob's arrow snaps onto $|\psi\rangle$ only after the corrections.



Noise analysis. Noise is inserted as explicit error channels after every gate (`qrelay.noise.noisify`, recursing into the `if_else` blocks). Average fidelity is computed exactly from the PTM via $F_{avg} = (2F_{pro} + 1)/3$.



- With CX-only depolarizing noise, simulation and our closed form $F = (1 + (1-p)^2)/2$ (THEORY §1.3) agree to 10^{-6}. The **quantum-advantage threshold** is $p_2^* = 1 - 1/\sqrt{3} = 0.4226$.
- Adding 1-qubit noise ($p_1 = p_2/10$) lowers it to $p_2^* = 0.384$. Adding readout error ($p/5$) lowers it to 0.330. Readout matters because it corrupts **Alice's feed-forward bits**.
- Thermal relaxation (gate times 50/300 ns): $F_{avg} = 0.997$ at $T_1 = T_2 = 200\,\mu s$, 0.945 at $10\,\mu s$, and it crosses 2/3 near $1\,\mu s$.
- **Readout mitigation** (confusion-matrix inversion) on the noisy model (p_1=0.01, p_2=0.1, readout 0.02): cardinal-state tomography $0.853 \to 0.868$. Mitigation cannot fix flipped feed-forward bits. For $|+\rangle$ with readout flip p the mitigated ceiling is exactly $1 - p$ (measured 0.924 at $p = 0.08$).



Real hardware [HW, IBM ibm_marrakesh (156-qubit Heron), job `dbgnmpote5oc738iia1g`, 4096 shots, cardinal-state tomography]. Deferred-measurement circuit: $F_{avg} = 0.978$ (per state 0.958–0.984, PTM diagonal 0.95/0.97/0.95). Dynamic circuit with real mid-circuit measurement and feed-forward: $F_{avg} = 0.880$. The $|0angle, |1angle$ states stay at 0.96–0.98, but the equatorial states drop to about 0.82, and the PTM shows X/Y = 0.65/0.69 against Z = 0.94. That is the signature of **dephasing while the qubit waits for the classical feed-forward**. Both circuits are far above 2/3: **genuine**

quantum teleportation on hardware. The whole batch used 55 s of QPU time. For comparison, the emulated device (EMU, fake_brisbane) gave 0.965 / 0.962.

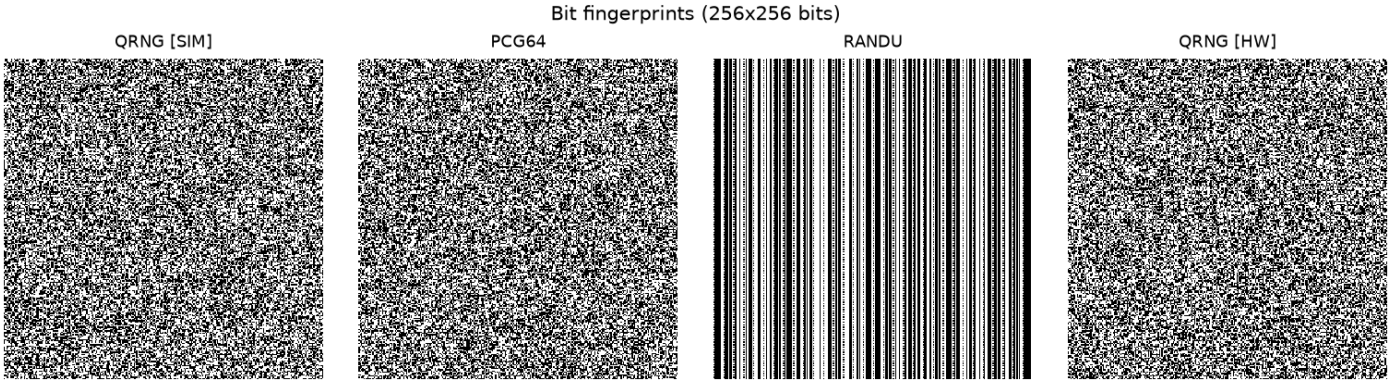
2. PS2: Quantum random number generator

Approach. H then measure, on 16 parallel qubits per shot (a 1-qubit textbook circuit and an m-qubit version). **1,000,000 bits** generated on the simulator in about 2 s, and **819,200 physically random bits** from IBM ibm_marrakesh (100 qubits × 8192 shots). Little-endian ordering is handled explicitly (`qrng.memory_to_bits`).

Statistics. 10 NIST SP 800-22-style tests implemented from the specification ($\alpha = 0.01$), plus Shannon and min-entropy:

generator (1M bits)	passed	min-entropy / byte	notes
QRNG [SIM]	**10/10**	7.807	lowest p = 0.015 (runs)
PCG64	10/10	7.828	
Mersenne Twister	10/10	7.828	
RANDU (low byte)	**4/10**	5.999	fails longest-run, serial, ApEn, DFT, autocorrelation, byte χ^2 with p < 1e-4
QRNG [HW] ibm_marrakesh (819,200 bits)	**8/10**	7.811	fails monobit and cumulative sums (p = 0.002): a real per-qubit bias, P(1) from 0.435 to 0.530
QRNG [EMU] (163,840 bits)	10/10	7.489	smaller sample, so lower plug-in min-entropy

RANDU is the control that proves the suite discriminates. Its **bit fingerprint** shows stripes where the others look like noise:



Interpretation. The quantum and classical generators are statistically indistinguishable by these tests, which is expected, because the tests cannot detect *where* randomness comes from. The value of a QRNG is the physical unpredictability of **hardware** bits, which no test can certify. The simulator's bits are, honestly, PRNG output.

Bias mitigation on a physically motivated biased source (asymmetric readout $P(0|1) = 0.2$, so $P(1) = 0.399$, $H_\infty = 0.735$/bit), 400k bits:

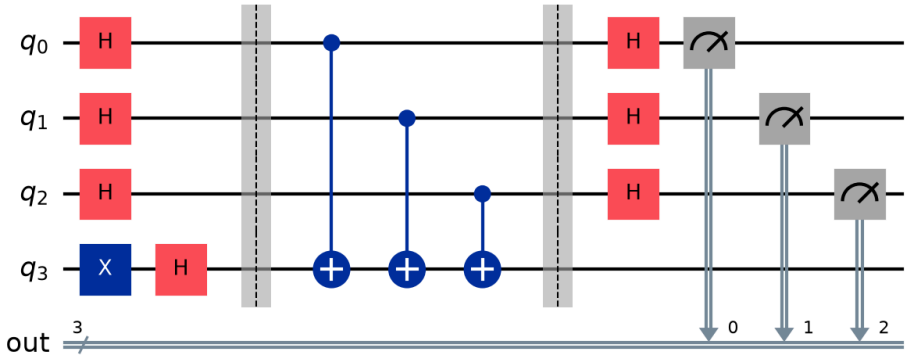
method	P(1) out	yield	tests
raw	0.3993	100%	0/10
Von Neumann	0.5011	23.9%	10/10
XOR pairs	0.4789	50%	2/10 (bias $0.1 \to 0.02 = 2\varepsilon^2$, as predicted)
Toeplitz hashing ($n = 1024$, $\varepsilon = 2^{-32}$)	0.4989	67.2%	10/10

Toeplitz recovers nearly the source's min-entropy, about 3× Von Neumann's yield, at the cost of an independent seed. **On the real hardware bits** ($P(1) = 0.5017$, 8/10 tests), every extractor restores 10/10: Von Neumann keeps 25%, XOR

50%, and Toeplitz 93% (763,200 bits). The first 200k hardware bits alone pass all tests; the bias only becomes detectable with the full sample.

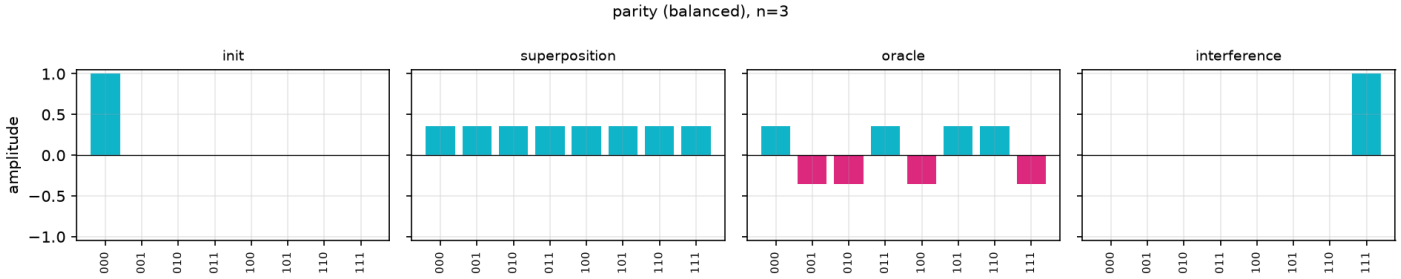
3. PS3: Deutsch–Jozsa

Approach. General n (tested 1–8). Oracles: 2 constant (identity, X), and balanced x_0, parity, inner product $a \cdot x$ and **random balanced** (a random half of the inputs, via MCX), plus an equivalent `Diagonal` phase oracle that tests show matches the bit oracle on every input. Random oracles draw from the QRNG pool.

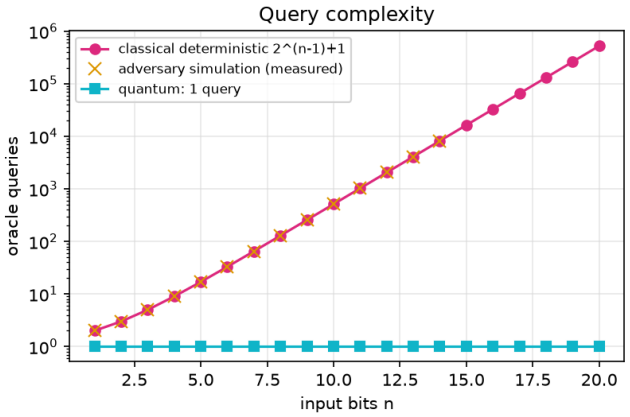


Correctness. We enumerate **every** function for $n = 1, 2, 3$ (2 constant and 2, 6, 70 balanced) and confirm $P(0^n) = 1$ or 0 exactly. We also test 300 random balanced oracles each for $n = 4, 5, 6$ and 40 each for $n = 7, 8$. Measured runs give all-zeros for constant and non-zero for balanced in every case.

Superposition, phase kickback, interference. The step view shows uniform amplitudes $1/\sqrt{2^n}$, then signs $(-1)^{f(x)}$ appearing after the single query while the ancilla stays $|-\rangle$, then the $|0^n\rangle$ amplitude reaching ± 1 (constant) or cancelling to exactly 0 (balanced).

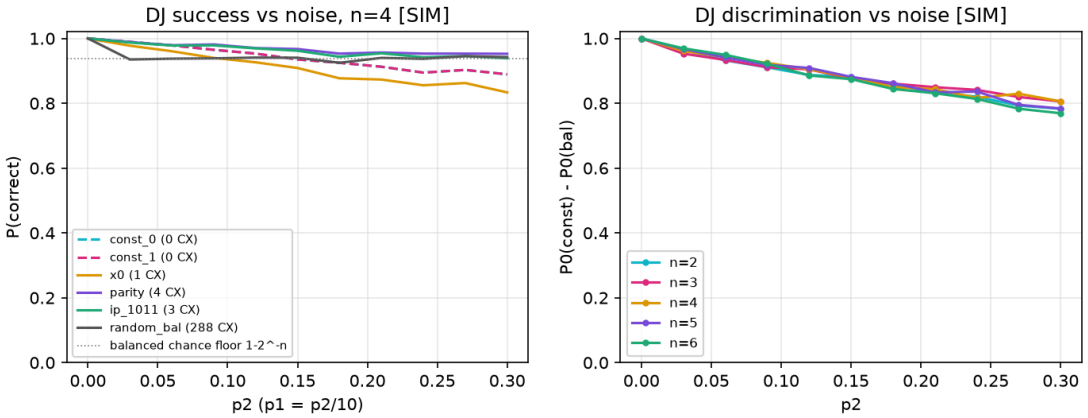


Query complexity. Quantum 1 versus classical deterministic $2^{n-1} + 1$. An **adversary simulation** (answer 0 for as long as possible) measures exactly $2^{n-1} + 1$ for $n = 1..14$.



Noise. Success degrades with the oracle's entangling-gate count. At $p_2 = 0.15$, $n = 2$: parity (2 CX) scores 0.962 and random balanced (12 CX) scores 0.794. **Pitfall we flag:** at large n the random-balanced oracles (288–3904 CX) *appear*

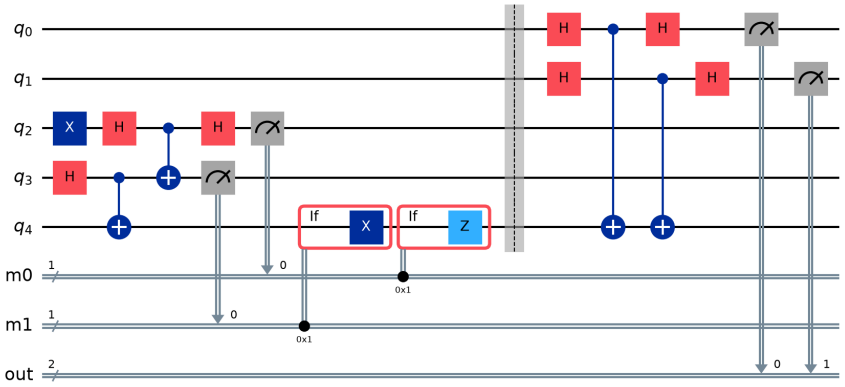
to succeed 94–98% under noise, but only because the output is nearly uniform and $P(\text{non-zero}) \to 1 - 2^{-n}$ by chance. The discrimination $P_0(\text{const}) - P_0(\text{bal})$ removes this artefact and decays monotonically.



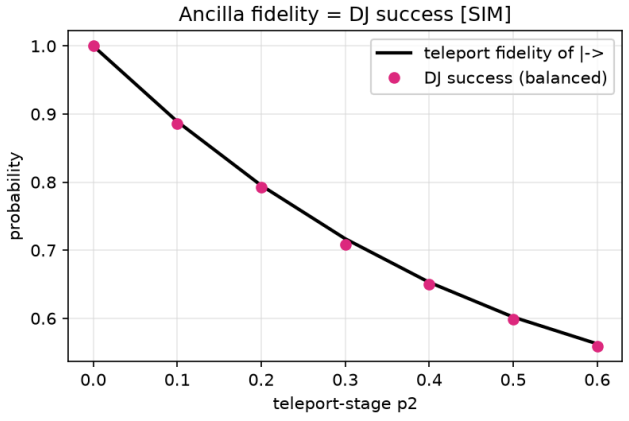
Real hardware [HW, ibm_marrakesh]. All 10 circuits are classified correctly. Success is 0.98–0.998 for constant and parity oracles, and 0.888 (n=2, 19 two-qubit gates) and 0.855 (n=3, 75 two-qubit gates) for random balanced oracles. The **teleported-ancilla distributed DJ** ran as one dynamic circuit on hardware: 0.996 (constant) and 0.861 (parity). EMU predicted the same ordering (0.95 and 0.89 for random balanced).

4. The integrated mission: teleported-ancilla Deutsch–Jozsa

The QRNG pool seeds a Haar-random state, the DJ size n and a hidden oracle. Alice teleports the state (Stage 2). Then she teleports $|-\rangle$, which **Bob uses as the DJ ancilla** (Stage 3, one dynamic circuit). One noise knob drives all stages.

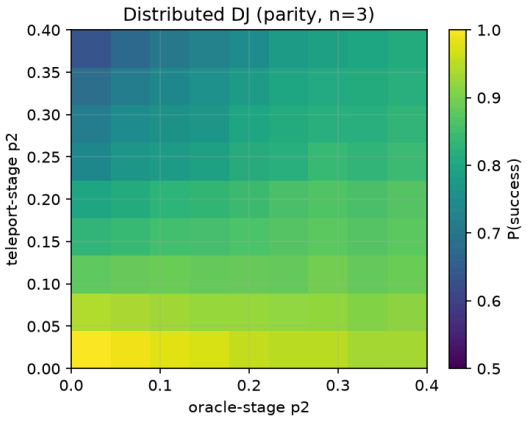


Result (proved in THEORY §4, verified numerically). With a noiseless oracle, constant oracles are always correct, and $P(\text{correct} \mid \text{balanced}) = \langle -|\rho_{anc}|- \rangle$, **the teleportation fidelity of the ancilla**. A $|+\rangle$ admixture gives no kickback, so teleportation noise makes Bob over-report "constant". Across $p_2 \in [0, 0.6]$ the agreement is within 0.009 (shot noise).



Which stage hurts more? For a shallow oracle (parity, n=3) teleport-stage noise dominates: $p_2 = 0.4$ gives 0.636 when confined to teleportation versus 0.933 when confined to the oracle. In the ideal case, the teleported ancilla is identical

to a local one for every library oracle ($n \leq 4$, both variants).



5. Testing, limitations, future work

Tests (53, all passing). Exact teleport fidelity (all branches, 50 random states, both variants), PTM = I, analytic noise formula, exhaustive DJ $n \leq 3$, bit/phase oracle equivalence, the adversary count, extractor improvement, monotone fidelity and the 2/3 crossing, the mitigation ceiling, the ancilla-fidelity theorem, mission event order, and every API endpoint plus the WebSocket. The UI is smoke-tested headlessly with zero console errors.

Limitations. SIM randomness is pseudo-random. EMU uses a calibration snapshot, not today's device. The Toeplitz extractor's guarantee assumes the per-bit min-entropy estimate holds, which is plug-in and i.i.d. assumed. The tomography projection onto the Bloch ball is a simple constrained estimate, not full maximum likelihood.

Future work. Mitigate the feed-forward dephasing (dynamical decoupling during the classical wait). Add Pauli twirling and dynamical decoupling to the teleportation circuit. Use entanglement-swapping relays (multi-hop Q-Relay). Add device-independent randomness certification via a CHSH test.

References: see `docs/REFERENCES.md`.